

Microscopic Condensation in Exponential-Feedback Transseries

Sharp coefficients, Gaussian–Poisson limits,
finite-perturbation universality, and refined Borel growth

Research draft prepared for Vladimir Reshetnikov

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Abstract

Consider the formal fixed point $\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(\lambda_j \widehat{U}(q))$, with nonnegative slopes. The ProveIt research article on this model leaves open a full multiplicative coefficient equivalent and correctly centered fluctuation laws beyond its logarithmic asymptotic. We resolve that question for an asymptotically quadratic class, including every finite perturbation of $\lambda_j = j^2$. If $\lambda_j = j^2 + \alpha j + \eta j / \log j + o(j / \log j)$, $b = 1 + \lambda_1$, and $r_n(r_n + 1)e^{2r_n} = n$, then

$$[q^n] \widehat{U}(q) \sim \frac{\exp\left((2r_n + 1) \frac{nr_n}{r_n + 1} + br_n^2 + \alpha r_n + \eta/2\right)}{\sqrt{2r_n^2 + 4r_n + 1}}.$$

The proof identifies the full microscopic configuration: one large action, a Poisson number of action-two factors, and otherwise action-one factors. The large action has Gaussian fluctuations, asymptotically independent of the Poisson cloud. This yields a three-level universality law, a multiplicative positive-axis Borel-growth equivalent, and an equivalent for the least term. In particular, changing only the first slope changes coefficients by a lognormal-size factor, although it leaves the previously known logarithmic asymptotic and Gevrey classification unchanged. All main results are proved here by positive Lagrange weights and two discrete saddle analyses. Exact finite computations accompany, but do not replace, the proofs.

Editorial note (ProveIt, 2026-09-29). This article was written at a snapshot that did not yet contain the package `Finite_Core_Universality_Exponential_Feedback/` [6] in `Analysis/Transseries/docs/series-and-transseries/`, filed before it (batch 47), and it does not cite that package. For slopes that are eventually exactly j^2 , [theorem 2.1](#) is the $p = 2$, $a = 1$ case of that package’s first-correction formula (its Proposition `prop:first-correction`, with $\lambda_1 + w_2 = b$), the variance $2n/(\log n)^2$ is its equation `eq:variance`, and the least-term index of [theorem 2.6](#) is its Corollary `cor:least-term`; that package also proves a local Gaussian law and a central limit theorem for the largest action (its Theorem `thm:clt`). The two are independent proofs of the same coefficient theorem, and their constants agree. New here are the αj and $\eta j / \log j$ terms, the rate (2.7), the joint total-variation Gaussian–Poisson law of the largest part and the number of twos, which answers that package’s question on joint laws for the finite background cloud at $p = 2$, $a = 1$, the two-dimensional local limit, the Borel equivalent and the value of the least term. The total-variation Poisson law at diverging intensity is also what the later package `Poisson_Layers_Finite_Core_Boundary/` lists as open; this article proves it for this class. Neither package has been independently reviewed.

Scope and status. This is a mathematical research draft, not a Lean verification or a claim of settling a named longstanding conjecture. It strengthens the specific repository article identified in [section 1](#). General publication priority has not been established. The least-term result is not asserted to be the remainder of an analytically summed solution.

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1 Repository context and the question being resolved

The relevant model is developed in ProveIt’s *A Sharp Regularity Classification for Countable Exponential-Feedback Transseries* [1]. The inspected repository snapshot is

04473354a0f3edff2d6c365caf26170ca6b88735.

The article lies under

Analysis/Transseries/docs/series-and-transseries/
Exponential_Feedback_Regularity_Classification/article.tex.

Its subsection *Beyond logarithmic coefficient asymptotics* explicitly asks for sublinear terms, a full multiplicative equivalent, and a local limit law for the number of factors and the largest action. It points out that small nonunit actions and the displacement of the large-action saddle become relevant at that precision. Those are exactly the effects analyzed here.

For the quadratic model the previous logarithmic result is

$$\log u_n = n \log n - 2n \log \log n + (\log 4 - 1)n + o(n). \quad (1.1)$$

The previous concentration results locate the largest action at $2n/\log n$, up to a relative error tending to zero, and the previous Borel result is $\log \log B(t) \sim 2\sqrt{t}$. None of these assertions determines a multiplicative equivalent. In particular, the factor $e^{2r_n^2}$ obtained below is invisible in (1.1), yet grows faster than every fixed power of n .

The broader transseries volume supplies the formal-versus-analytic context [2]. Its own status inventory distinguishes theorems with precise Lean counterparts from manuscript-level results. The present article makes no status promotion for either that volume or the research package. Only selected, relevant repository documents were inspected; this is not an exhaustive audit of the repository.

1.1 Relationship to standard methods

The positive coefficient formula is classical Lagrange inversion, for which Gessel’s survey is a convenient reference [3]. The idea of condensation—one component carrying an unusually large share of a total—has a substantial literature; Janson treats it for simply generated trees and random allocations [4]. We do not claim to introduce condensation, Lagrange inversion, Poisson approximation, or the saddle-point method.

The model here has the non-product weight $(\sum_j \lambda_j m_j)^{\sum_j m_j - 1} / \prod_j m_j!$. Its large action is of order $n/\log n$, not of order n , and its microscopic cloud has logarithmically growing size. The explicit equivalent, the one-large-action/ones/twos reduction, and the resulting universality constants are the model-specific refinements proved below. All necessary estimates are supplied, rather than importing a random-allocation theorem whose hypotheses have not been checked.

2 Model, assumptions, and main theorems

Let $\lambda_j \geq 0$ be finite real numbers. We work first in $\mathbb{R}[[q]]$, with

$$\widehat{U}(q) = \sum_{j \geq 1} q^j e^{\lambda_j \widehat{U}(q)} = \sum_{n \geq 1} u_n q^n. \quad (2.1)$$

The identity is formal: no analytic convergence near $(q, u) = (0, 0)$ is presupposed. On writing $q = e^{-x}$, it becomes a one-grid exponential transseries $\sum_n u_n e^{-nx}$. Well-based support and growth of its coefficients are different questions.

The principal asymptotic hypothesis is

$$\lambda_j = j^2 + \alpha j + \eta \frac{j}{\log j} + o\left(\frac{j}{\log j}\right) \quad (j \rightarrow \infty), \quad \lambda_j \geq 0. \quad (\text{Q})$$

The expression involving $\log j$ is used only on a tail. The finitely many initial slopes, especially λ_1 , are arbitrary nonnegative numbers. Set

$$b = 1 + \lambda_1, \quad D(r) = 2r^2 + 4r + 1, \quad r_n(r_n + 1)e^{2r_n} = n, \quad (2.2)$$

where the last equation has one positive real solution. Define

$$x_n = \frac{n}{r_n + 1}, \quad m_n = \frac{nr_n}{r_n + 1}, \quad \sigma_n^2 = \frac{m_n}{D(r_n)}. \quad (2.3)$$

For later comparison,

$$r_n = \frac{1}{2} \log n - \log \log n + \log 2 + O\left(\frac{\log \log n}{\log n}\right), \quad (2.4)$$

$$x_n \sim \frac{2n}{\log n}, \quad \sigma_n^2 \sim \frac{2n}{(\log n)^2}. \quad (2.5)$$

The implicit definition of r_n , not a finite truncation of (2.4), is to be used in the multiplicative formulas.

Theorem 2.1 (Full coefficient equivalent). *Under (Q), the formal solution exists uniquely, has $u_1 = 1$ and $u_n \geq 1$, and*

$$u_n \sim F_n := \frac{\exp((2r_n + 1)m_n + br_n^2 + \alpha r_n + \eta/2)}{\sqrt{D(r_n)}}. \quad (2.6)$$

If $\lambda_j = j^2$ for all sufficiently large j , then $\alpha = \eta = 0$ and the stronger bound

$$u_n = F_n \left(1 + O_\lambda \left(\frac{(\log n)^2}{\sqrt{n}} \right) \right) \quad (2.7)$$

is valid. The constant may depend on the fixed exceptional slopes.

To state the probabilistic refinement, let

$$\mathcal{M}_n = \left\{ \mathbf{m} = (m_1, m_2, \dots) : m_j \in \mathbb{N}, \sum_j j m_j = n \right\}. \quad (2.8)$$

Here $\mathbb{N}$ includes zero and only finitely many entries are nonzero. Put

$$K = \sum_j m_j, \quad S = \sum_j \lambda_j m_j, \quad W(\mathbf{m}) = \frac{S^{K-1}}{\prod_j m_j!}, \quad \mathbb{P}_n(\mathbf{m}) = \frac{W(\mathbf{m})}{u_n}, \quad (2.9)$$

with $0^0 = 1$. The coefficient formula below proves that this is a probability distribution. Write $J = \max\{j : m_j > 0\}$ and $D_2 = m_2$.

Definition 2.2. The discrete Gaussian G_n has law on $\mathbb{Z}$

$$\mathbb{P}(G_n = k) = \frac{1}{Z_n} \exp\left(-\frac{(k - x_n)^2}{2\sigma_n^2}\right), \quad Z_n = \sum_{k \in \mathbb{Z}} \exp\left(-\frac{(k - x_n)^2}{2\sigma_n^2}\right). \quad (2.10)$$

Let P_n be independent of G_n , with distribution $\text{Pois}(r_n^2)$.

Theorem 2.3 (Microscopic condensation and joint local approximation). *Under (Q), with probability*

$$1 - O_\lambda\left(\frac{(\log n)^4}{n}\right), \quad (2.11)$$

there is exactly one part of size at least three. Its size is J , and all remaining parts have sizes one or two. Moreover,

$$d_{\text{TV}}(\mathcal{L}_n(J, D_2), \mathcal{L}(G_n, P_n)) \rightarrow 0. \quad (2.12)$$

For eventually exact quadratic slopes the distance in (2.12) is $O_\lambda((\log n)^2/\sqrt{n})$. Consequently,

$$\left(\frac{J - x_n}{\sigma_n}, \frac{D_2 - r_n^2}{r_n}\right) \Rightarrow (Z_1, Z_2), \quad (2.13)$$

where Z_1, Z_2 are independent standard normal random variables.

Thus a Poisson law, not merely a negligible relative error, describes the small nonunit cloud. Its mean is asymptotic to $(\log n)^2/4$, while the large-action variance is asymptotic to $2n/(\log n)^2$.

Corollary 2.4 (A three-level universality law). *Let two nonnegative slope sequences satisfy (Q), with parameters $(\lambda_1, \alpha, \eta)$ and $(\tilde{\lambda}_1, \tilde{\alpha}, \tilde{\eta})$. Then*

$$\frac{u_n}{\tilde{u}_n} \sim \exp\left((\lambda_1 - \tilde{\lambda}_1)r_n^2 + (\alpha - \tilde{\alpha})r_n + \frac{\eta - \tilde{\eta}}{2}\right). \quad (2.14)$$

In particular, finite changes to slopes with indices $j \geq 2$ do not alter the multiplicative equivalent when the first slope is held fixed. A change of the first slope does alter it, by the lognormal-size factor $e^{(\lambda_1 - \tilde{\lambda}_1)r_n^2}$.

Theorem 2.5 (Refined positive-axis Borel growth). *The integrated ordinary Borel transform*

$$B(t) = \sum_{n \geq 1} \frac{u_n}{n!} t^n \quad (2.15)$$

is entire. Put $s = s(t) = (\sqrt{1 + 4t} - 1)/2$, so $s(s + 1) = t$. As $t \rightarrow +\infty$,

$$B(t) \sim \frac{\exp(se^{2s} + bs^2 + \alpha s + \eta/2)}{\sqrt{2s + 1}}. \quad (2.16)$$

This is a multiplicative equivalent, not only an equivalent for a logarithm or a double logarithm.

Theorem 2.6 (Least-term equivalent). *For $t > 0$ let $L(t) = \min_{n \geq 1} u_n t^n$. As $t \downarrow 0$, put $R = \frac{1}{2} \log(1/t)$. Then*

$$L(t) \sim \frac{\exp(-R^2/t + bR^2 + \alpha R + \eta/2)}{\sqrt{2R^2 + 4R + 1}}. \quad (2.17)$$

Every minimizing index is asymptotic to

$$N_0(t) = \frac{R(R + 1)}{t}. \quad (2.18)$$

Choosing a nearest integer to $N_0(t)$ already gives a term asymptotic to $L(t)$.

3 Formal solution and positive Lagrange weights

For $V \in q\mathbb{R}[[q]]$, every coefficient of $\sum_j q^j e^{\lambda_j V}$ receives contributions only from $j \leq n$. If $V - W$ has order $h \geq 1$, then $e^{\lambda_j V} - e^{\lambda_j W}$ has order at least h , and the outer factor q^j raises it to at least $h + 1$. Picard iteration therefore determines a unique formal solution coefficient by coefficient. Its first coefficient is one.

Lemma 3.1 (Coefficient identity). *For every $n \geq 1$,*

$$u_n = \sum_{k=1}^n \frac{1}{k!} \sum_{\substack{j_1 + \dots + j_k = n \\ j_i \geq 1}} (\lambda_{j_1} + \dots + \lambda_{j_k})^{k-1} \quad (3.1)$$

$$= \sum_{m \in \mathcal{M}_n} \frac{(\sum_j \lambda_j m_j) \sum_j m_j - 1}{\prod_j m_j!}. \quad (3.2)$$

In particular all weights are nonnegative, and the $k = 1$ term gives $u_n \geq 1$.

Proof. Introduce an auxiliary variable z and solve $V = z\Phi(q, V)$, where $\Phi(q, v) = \sum_j q^j e^{\lambda_j v}$. Formal Lagrange inversion gives

$$V = \sum_{k \geq 1} \frac{z^k}{k!} \left. \frac{\partial^{k-1}}{\partial v^{k-1}} \Phi(q, v)^k \right|_{v=0}.$$

At coefficient q^n , only $k \leq n$ occurs, so setting $z = 1$ is legitimate coefficientwise. Differentiating the product of exponentials produces (3.1). Grouping ordered compositions by their multiplicities cancels the factor $k!$ and yields (3.2). $\square$

For the unperturbed quadratic sequence the first coefficients are

$$u_1 = 1, \quad u_2 = 2, \quad u_3 = \frac{15}{2}, \quad u_4 = \frac{107}{3}, \quad u_5 = \frac{1555}{8}, \quad u_6 = \frac{17362}{15}. \quad (3.3)$$

In general $u_2 = 1 + \lambda_1 = b$. Its reappearance as the coefficient of r_n^2 in the large-order equivalent is a concrete link between the first nonlinear coefficient and the distant tail.

4 Coarse localization, proved at exponential precision

This section supplies the localization needed for a self-contained proof. It is a quadratic specialization of the envelope mechanism in the source article, but no unproved concentration statement is needed subsequently. Write $\ell = \log n$ in estimates as $n \rightarrow \infty$.

Lemma 4.1 (Coarse large-action localization). *Suppose $\lambda_j = j^2 + O(j)$, with nonnegative slopes. With $R_n = n - K + 1$, every fixed $\epsilon > 0$ satisfies*

$$\mathbb{P}_n \left(\left| \frac{R_n \ell}{n} - 2 \right| > \epsilon \right) \leq e^{-c_\epsilon n + o(n)}. \quad (4.1)$$

For every fixed $\delta > 0$,

$$\mathbb{P}_n(J \leq (1 - \delta)R_n) \leq e^{-c_\delta n + o(n)}. \quad (4.2)$$

In particular, for any fixed $1 < a < 2 < c < \infty$,

$$\mathbb{P}_n(J \notin \mathcal{I}_n) \leq e^{-c_0 n + o(n)}, \quad \mathcal{I}_n = \left[\frac{an}{\ell}, \frac{cn}{\ell} \right] \cap \mathbb{Z}. \quad (4.3)$$

Proof. Choose C so that $\lambda_j \leq j^2 + Cj$ for all j . For a composition with k parts, put $R = n - k + 1$. If $d_i = j_i - 1$, then $\sum_i d_i = R - 1$ and

$$\sum_i j_i^2 = \sum_i d_i^2 + 2(R - 1) + k \leq R^2 + k - 1.$$

Thus the sum of slopes is at most $R^2 + C_1 n$, where C_1 is fixed. The entire row with $k = n - R + 1$ parts is consequently bounded by

$$T_n(R) = \binom{n-1}{R-1} \frac{(R^2 + C_1 n)^{n-R}}{(n-R+1)!}. \quad (4.4)$$

A lower bound is obtained from one large part $J = \lfloor 2n/\ell \rfloor$ and $n - J$ ones:

$$u_n \geq \frac{\lambda_J^{n-J}}{(n-J)!}.$$

Since $\lambda_J = J^2(1 + O(1/J))$, Stirling's formula gives

$$\log u_n \geq n\ell - 2n \log \ell + (\log 4 - 1)n + o(n). \quad (4.5)$$

For completeness, the maximization of (4.4) can be localized without losing a constant of order n . If $R \geq \rho n$ for a fixed $\rho > 0$, its logarithm is at most $(1 - \rho)n\ell + O(n \log \ell) + O(n)$, which is far below (4.5). A row within $O(n)$ of that lower bound must therefore have $R/n \rightarrow 0$. Its binomial entropy is then $o(n)$. The same comparison forces $\log(R^2 + C_1 n) \geq 2\ell - O(\log \ell)$, so $R^2/n \rightarrow \infty$, $\log R = \ell - O(\log \ell)$, and the additive $C_1 n$ is negligible. A further comparison gives $(R/n)\ell = O(\log \ell)$.

Put $y = R\ell/n$. Uniform Stirling estimates in the just-localized rows now yield

$$\frac{1}{n} \log T_n(R) = \ell - 2 \log \ell + 1 + 2 \log y - y + o(1). \quad (4.6)$$

The function $2 \log y - y$ tends to minus infinity at either endpoint of $(0, \infty)$ and has its unique maximum at $y = 2$. On $|y - 2| \geq \epsilon$ it loses a fixed positive amount. The preliminary localization shows that rows not covered by (4.6) lose at least as much: otherwise a sequence of competing rows would have satisfied those localization conditions. Summing at most n rows changes a logarithm by at most $\log n = o(n)$. This proves (4.1), and also the matching upper version of (4.5).

For the largest part, always $J \leq R$. If $J \leq (1 - \delta)R$, the stronger estimate $\sum_i d_i^2 \leq (J - 1)(R - 1)$ gives

$$\sum_i \lambda_{j_i} \leq JR + C_2 n.$$

On the already localized rows, $R^2 \gg n$, so the right side is at most $(1 - \delta/2)R^2$ for large n . Replacing the numerator of (4.4) by this capped value loses a factor $(1 - \delta/2)^{n-R}$, hence e^{-cn} . The binomial entropy on those rows is only $e^{o(n)}$; it cannot cancel that loss. This proves (4.2). Combining the two estimates, with slightly narrower windows for R , proves (4.3). $\square$

5 The microscopic row estimate

Fix $J \in \mathcal{I}_n$, and write

$$m = n - J, \quad \Lambda = \lambda_J, \quad t_J = \frac{m}{\Lambda}, \quad A_J = \frac{m^2}{\Lambda}, \quad C_J = \frac{\Lambda^m}{m!}. \quad (5.1)$$

In this section assume only $\lambda_j = j^2 + O(j)$. Uniformly on the window, $\Lambda \sim J^2$, $t_J \asymp \ell^2/n$, and $A_J \asymp \ell^2$. Let $u_{n,J}$ denote the sum of the weights with largest part exactly J .

Lemma 5.1 (Uniform row equivalent and microscopic law). *Uniformly for $J \in \mathcal{I}_n$,*

$$u_{n,J} = C_J e^{bA_J} \left(1 + O_\lambda \left(\frac{\ell^4}{n} \right) \right). \quad (5.2)$$

Conditionally on the largest part being J , the probability that there is any additional part of size at least three is $O_\lambda(\ell^4/n)$. The conditional distribution of D_2 is at that same total-variation distance from $\text{Pois}(A_J)$.

5.1 An upper bound that keeps the small-part entropy

Distinguish one occurrence of the largest part J . Let a_h be the remaining multiplicities, so $\sum_{h \leq J} h a_h = m$. Write $r = \sum_h a_h$ and $s = \sum_h \lambda_h a_h$. The weight with the largest part marked is

$$\frac{(\Lambda + s)^r}{\prod_{h \leq J} a_h!}. \quad (5.3)$$

If there are several occurrences of J , marking overcounts the original weight by their number. Therefore a sum of marked weights is a valid upper bound; when all remaining parts are ones or twos, it is exact.

Because $s \geq 0$ and $r \leq m$,

$$(\Lambda + s)^r \leq \Lambda^r \exp(ms/\Lambda).$$

Use

$$a_1 = m - \sum_{h \geq 2} h a_h, \quad r = m - \sum_{h \geq 2} (h-1) a_h, \quad s = \lambda_1 m + \sum_{h \geq 2} (\lambda_h - \lambda_1 h) a_h,$$

and $m!/a_1! \leq m^{m-a_1}$. Dividing (5.3) by C_J gives the upper bound

$$e^{\lambda_1 A_J} \prod_{h=2}^J \frac{\mu_h^{a_h}}{a_h!}, \quad \mu_h = \frac{m^h}{\Lambda^{h-1}} \exp(t_J(\lambda_h - \lambda_1 h)). \quad (5.4)$$

We may drop the remaining mass constraint when summing a nonnegative upper bound. Hence

$$u_{n,J} \leq C_J \exp \left(\lambda_1 A_J + \sum_{h=2}^J \mu_h \right). \quad (5.5)$$

Lemma 5.2 (Suppression of parts of size at least three). *Uniformly on $\mathcal{I}_n$,*

$$\mu_2 = A_J + O_\lambda(\ell^4/n), \quad \sum_{h=3}^J \mu_h = O_\lambda(\ell^4/n). \quad (5.6)$$

Proof. The first claim follows immediately from $\mu_2 = A_J e^{t_J(\lambda_2 - 2\lambda_1)}$. For the second choose a fixed C with $\lambda_h - \lambda_1 h \leq h^2 + Ch$ for all h , and bound μ_h by

$$v_h = \Lambda t_J^h e^{t_J(h^2 + Ch)}.$$

For $3 \leq h \leq J/2$,

$$\log(v_{h+1}/v_h) = \log t_J + t_J(2h + 1 + C) \leq -(1 - 1/a + o(1))\ell.$$

Thus that portion is geometrically decreasing and has sum $O(v_3) = O(\Lambda t_J^3) = O(\ell^4/n)$. For $J/2 \leq h \leq J$, the logarithm of v_h is a convex quadratic in h . At the endpoints its leading terms are, respectively, $-J\ell/2 + n/4 + o(n)$ and $-J\ell + n + o(n)$. Since $J\ell/n \geq a > 1$, both are at most $-c_1 n$. Convexity bounds the whole interval by the larger endpoint; its total contribution is $O(Je^{-c_1 n})$. This proves (5.6). $\square$

Combining lemma 5.2 with (5.5) proves the upper half of (5.2). It also bounds the marked mass with at least one additional part of size three or larger by

$$C_J e^{\lambda_1 A_J + \mu_2} \left(\exp \left(\sum_{h=3}^J \mu_h \right) - 1 \right) = C_J e^{b A_J} O_\lambda(\ell^4/n). \quad (5.7)$$

This bound includes configurations with a second largest part equal to J .

5.2 Matching lower bound and the Poisson cloud

Retain exactly one J , exactly d twos, and $m - 2d$ ones. Its unmarked weight is

$$w_{J,d} = \frac{(\Lambda + \lambda_1 m + (\lambda_2 - 2\lambda_1)d)^{m-d}}{(m-2d)!d!}, \quad 0 \leq d \leq m/2. \quad (5.8)$$

For any fixed sufficiently large C_3 , uniformly for $0 \leq d \leq C_3 \ell^2$, expansion of the logarithm and of the falling factorial gives

$$(m-d) \log \left(1 + \frac{\lambda_1 m + (\lambda_2 - 2\lambda_1)d}{\Lambda} \right) = \lambda_1 A_J + O_\lambda(\ell^4/n), \quad (5.9)$$

$$\frac{m!}{(m-2d)!m^{2d}} = 1 + O(\ell^4/n). \quad (5.10)$$

For example, the error sizes in (5.9) are bounded by constants times $dm/\Lambda + m^3/\Lambda^2 = O(\ell^4/n)$; the factorial error is $O(d^2/m) = O(\ell^4/n)$. Therefore

$$w_{J,d} = C_J e^{\lambda_1 A_J} \frac{A_J^d}{d!} \left(1 + O_\lambda(\ell^4/n) \right). \quad (5.11)$$

Choose C_3 larger than twice the largest asymptotic coefficient in A_J/ℓ^2 . The upper tail of $\text{Pois}(A_J)$ past $C_3 \ell^2$ is $e^{-c_2 \ell^2}$, by the elementary exponential Markov bound $\mathbb{P}(P \geq k) \leq \inf_{z>1} e^{A_J(z-1)} z^{-k}$. That is smaller than ℓ^4/n . Summing (5.11) proves the matching lower bound in (5.2).

The same estimates, before summation, compare the normalized probabilities on the cutoff with the Poisson probabilities in total variation. The cutoff tail and (5.7) account for all remaining mass. This proves every assertion of lemma 5.1.

Remark 5.3 (Where the two contributions to b come from). The factor $e^{\lambda_1 A_J}$ is the feedback of the many unit-action factors. The factor e^{A_J} is their combinatorial replacement by twos. Dropping the latter is not an innocuous simplification: it would lose a factor of order $e^{r_n^2}$. The numerical value of λ_2 enters only the smaller error at this level of precision.

6 The large-action saddle and the full equivalent

The microscopic estimate reduces the original sum over all integer partitions to one positive sum over the largest action. We first analyze the case of eventually exact quadratic slopes; then we show precisely how the two additional tail parameters enter.

6.1 An exact saddle coordinate

For real $0 < x < n$, write

$$H_n(x) = (n-x) \left[1 + \log \frac{x^2}{n-x} \right], \quad A_0(x) = \left(\frac{n-x}{x} \right)^2. \quad (6.1)$$

Direct differentiation gives

$$H'_n(x) = \log \frac{n-x}{x^2} + \frac{2(n-x)}{x}, \quad (6.2)$$

$$H''_n(x) = -\frac{1}{n-x} - \frac{4}{x} - \frac{2(n-x)}{x^2} < 0, \quad (6.3)$$

$$H'''_n(x) = -\frac{1}{(n-x)^2} + \frac{6}{x^2} + \frac{4(n-x)}{x^3}. \quad (6.4)$$

There is exactly one critical point. Set $r = (n-x)/x$ there. Equation (6.2) becomes $x = re^{2r}$, equivalently $n = r(r+1)e^{2r}$. Thus the critical point is exactly x_n , and

$$H_n(x_n) = (2r_n + 1)m_n, \quad -H''_n(x_n) = \frac{D(r_n)}{m_n} = \sigma_n^{-2}. \quad (6.5)$$

Taking logarithms in the defining equation for r_n first gives $r_n \sim \frac{1}{2} \log n$. Substitution back into $2r_n = \log n - \log r_n - \log(r_n + 1)$ gives (2.4) and then (2.5).

6.2 A quantitative discrete Gaussian comparison

For $J \in \mathcal{I}_n$, Stirling's formula and lemma 5.1 give, when the slopes are eventually exactly quadratic,

$$u_{n,J} = e^{H_n(J)} a_n(J) \left(1 + O_\lambda(\ell^4/n)\right), \quad a_n(x) = \frac{e^{bA_0(x)}}{\sqrt{2\pi(n-x)}}. \quad (6.6)$$

Here and below the fixed macroscopic interval may be enlarged slightly to contain each real segment between J and x_n . Uniformly on that interval,

$$-H''_n(x) \asymp \frac{\ell^2}{n}, \quad |H'''_n(x)| = O(\ell^3/n^2), \quad |(\log a_n)'(x)| = O_\lambda(\ell^3/n). \quad (6.7)$$

The last estimate follows from

$$(\log a_n)'(x) = -\frac{2bn(n-x)}{x^3} + \frac{1}{2(n-x)}.$$

Let $y = (J - x_n)/\sigma_n$ and $\varepsilon_n = \ell^2/\sqrt{n}$. Taylor's theorem, strict concavity, and (6.7) imply

$$H_n(J) - H_n(x_n) = -\frac{1}{2}y^2 + O(|y|^3/\sqrt{n}), \quad (6.8)$$

$$H_n(J) - H_n(x_n) \leq -c_1 y^2, \quad (6.9)$$

$$\left| \log \frac{a_n(J)}{a_n(x_n)} \right| \leq C_1 \varepsilon_n |y|. \quad (6.10)$$

These statements hold throughout the fixed macroscopic interval, not just at a fixed value of y .

To make the error after summation explicit, use $|e^v - e^w| \leq |v - w|e^{\max(v,w)}$. Equations (6.8)–(6.10) then bound the difference of the scaled summand and $e^{-y^2/2}$ by

$$Ce^{-c_2 y^2 + C_2 \varepsilon_n |y|} \left(\frac{|y|^3}{\sqrt{n}} + \varepsilon_n |y| \right). \quad (6.11)$$

The lattice spacing in y is $\sigma_n^{-1} \rightarrow 0$. Gaussian moments, or a comparison on intervals of that length, show that the sum of (6.11) is $O_\lambda(\sigma_n \varepsilon_n)$. The elementary Riemann-sum estimate

$$\sum_{J \in \mathbb{Z}} e^{-(J-x_n)^2/(2\sigma_n^2)} = \sqrt{2\pi} \sigma_n + O(1) \quad (6.12)$$

is uniform in the fractional part of x_n . For instance, the error between the integral and the sum of a continuously differentiable function is bounded by the integral of the absolute derivative, which is bounded for this Gaussian. The Gaussian mass outside $\mathcal{I}_n$ is exponentially small. Since $1/\sigma_n = O(\ell/\sqrt{n}) = O(\varepsilon_n)$, we obtain

$$\sum_{J \in \mathcal{I}_n} u_{n,J} = e^{H_n(x_n)} a_n(x_n) \sqrt{2\pi} \sigma_n (1 + O_\lambda(\varepsilon_n)). \quad (6.13)$$

The row error $O(\ell^4/n) = O(\varepsilon_n^2)$ is smaller. Finally, [lemma 4.1](#) says that the left side is $u_n(1 + O(e^{-cn+o(n)}))$; no absolute estimate of omitted rows is being presumed. Substitution of (6.5) gives (2.6) and (2.7) in the eventually quadratic case.

6.3 Linear and critical sublinear tail perturbations

Assume now (Q), and let $\rho(x) = (n - x)/x$. Uniformly for $J \in \mathcal{I}_n$,

$$(n - J) \log \frac{\lambda_J}{J^2} = \alpha \rho(J) + \eta \frac{\rho(J)}{\log J} + o(1), \quad (6.14)$$

$$\frac{(n - J)^2}{\lambda_J} - A_0(J) = O_\lambda(\ell^3/n) = o(1). \quad (6.15)$$

For the first equality, the quadratic Taylor error is $O((n - J)/J^2) = O(\ell^2/n)$. The unspecified tail error, when multiplied by $(n - J)/J^2$, is $o((n - J)/(J \log J)) = o(1)$ uniformly, because all the indices in the window tend to infinity together and $\rho(J)/\log J$ is bounded. This uniformity is important: no regularity or differentiability of the unspecified o -term is assumed.

Consequently (6.6) holds with a relative $1 + o(1)$ error and with the smooth amplitude

$$a_n^*(x) = \frac{\exp(bA_0(x) + \alpha \rho(x) + \eta \rho(x)/\log x)}{\sqrt{2\pi(n - x)}}. \quad (6.16)$$

The additional logarithmic derivatives are $O_\lambda(\ell^2/n)$, so the bound (6.10) and the Gaussian comparison remain valid. At the saddle,

$$\rho(x_n) = r_n, \quad \log x_n = 2r_n + \log r_n, \quad \frac{r_n}{\log x_n} \longrightarrow \frac{1}{2}.$$

It follows that

$$a_n^*(x_n) = \frac{\exp(br_n^2 + \alpha r_n + \eta/2)}{\sqrt{2\pi m_n}} (1 + o(1)).$$

Repeating (6.13) proves [theorem 2.1](#) in full. For an arbitrary unspecified tail $o(j/\log j)$, its modulus of convergence contributes an unspecified $o(1)$; the quantitative rate (2.7) is therefore not asserted in that generality.

Remark 6.1 (Why the saddle must be specified accurately). An error small relative to x_n need not be small relative to σ_n . In particular, replacing x_n by $2n/\log n$ does not give a correctly centered Gaussian law. Similarly, substituting a short logarithmic expansion for r_n into the large exponent of (2.6) need not preserve a relative $1 + o(1)$ error. The implicit scalar equation eliminates this avoidable loss of precision.

7 The Gaussian–Poisson law and genuine local limits

The same estimates give distributional results without differentiating an asymptotic equivalent. Normalize (6.6) by the sum in (6.13). The summed absolute error (6.11), together with the

relative error in the row formula, proves

$$d_{\text{TV}}(\mathcal{L}_n(J), \mathcal{L}(G_n)) = o(1). \quad (7.1)$$

For eventually quadratic slopes the error is $O_\lambda(\varepsilon_n)$. The mass outside the macroscopic interval is already exponentially small.

7.1 Removing the dependence of the Poisson mean on the large action

Conditionally on $J \in \mathcal{I}_n$, [lemma 5.1](#) compares D_2 with $\text{Pois}(A_J)$, with total-variation error $O_\lambda(\ell^4/n)$. For two nonnegative means, the elementary Poisson coupling gives

$$d_{\text{TV}}(\text{Pois}(v), \text{Pois}(w)) \leq \min(1, |v - w|). \quad (7.2)$$

Indeed, when $v \leq w$, add an independent $\text{Pois}(w - v)$ variable; the resulting variables differ with probability at most $w - v$.

For $|J - x_n| = O(\sigma_n)$, equations [\(6.15\)](#) and $A'_0(x) = -2n(n - x)/x^3$ imply

$$A_J - r_n^2 = O_\lambda(\ell^2/\sqrt{n}) + o(1) = o(1). \quad (7.3)$$

In the eventually quadratic case, Gaussian domination from [\(6.9\)–\(6.10\)](#) gives the integrated bound

$$\mathbb{E}_n \left[\min(1, |A_J - r_n^2|); J \in \mathcal{I}_n \right] = O_\lambda(\varepsilon_n).$$

In the general case the bounded quantity in brackets tends to zero in probability, by [\(7.1\)](#) and [\(7.3\)](#); hence its expectation tends to zero. Mixture coupling, followed by [\(7.1\)](#), now proves [\(2.12\)](#), including its stated rate in the eventually quadratic case.

The probability of an extra part of size at least three is $O_\lambda(\ell^4/n)$ on every macroscopic row by [lemma 5.1](#). The complement of those rows has exponentially small probability. This proves [\(2.11\)](#). The standardized discrete Gaussian tends to a standard normal by [\(6.12\)](#). The characteristic function of $(P_n - r_n^2)/r_n$ is

$$\exp\left(r_n^2(e^{it/r_n} - 1 - it/r_n)\right) \longrightarrow e^{-t^2/2}.$$

Independence in the comparison law and vanishing total variation prove [\(2.13\)](#). This completes the proof of [theorem 2.3](#).

7.2 Point probabilities in the central window

Total-variation convergence alone is not a rescaled local limit theorem: a small total error could in principle be concentrated at a few lattice points. The available pointwise estimates rule that out here.

Corollary 7.1 (Central two-dimensional local limit). *Under [\(Q\)](#), for every fixed $M < \infty$, uniformly over integers j, d with $|j - x_n| \leq M\sigma_n$ and $|d - r_n^2| \leq Mr_n$,*

$$\sigma_n r_n \mathbb{P}_n(J = j, D_2 = d) = \frac{1}{2\pi} \exp\left(-\frac{(j - x_n)^2}{2\sigma_n^2} - \frac{(d - r_n^2)^2}{2r_n^2}\right) + o(1). \quad (7.4)$$

The same formula holds with the event replaced by $\{J = j, K = n + 1 - j - d\}$.

Proof. For configurations with only one part at least three, (5.11) is a uniform relative estimate at every displayed j, d . The pointwise saddle expansion and the full coefficient equivalent give the Gaussian factor. Stirling's formula for the Poisson probabilities, with $d = r_n^2 + O(r_n)$, gives the second Gaussian factor uniformly. Replacing A_j by r_n^2 has vanishing relative effect there, by (7.3).

For a fixed central j , all configurations with an additional part at least three have probability at most $O_\lambda(\ell^4/n)\mathbb{P}_n(J = j) = O_\lambda(\ell^4/(n\sigma_n))$. After multiplication by $\sigma_n r_n$, this is $O_\lambda(\ell^5/n) = o(1)$, even if all of that mass were at the particular d . Thus it cannot affect the limit. On the good configurations, $K = n + 1 - J - D_2$. The same bound handles bad configurations for the second event as well. $\square$

Corollary 7.2 (The factor count and the excess action). *Let $R_n = n - K + 1$. Under (Q),*

$$\frac{K - (n + 1 - x_n - r_n^2)}{\sigma_n} \implies -Z_1, \quad \frac{R_n - (x_n + r_n^2)}{\sigma_n} \implies Z_1. \quad (7.5)$$

Moreover, $R_n - J$ is at total-variation distance $o(1)$ from $\text{Pois}(r_n^2)$.

Proof. On the event of (2.11), $R_n = J + D_2$ exactly. Apply the joint approximation and use $r_n/\sigma_n \rightarrow 0$. $\square$

The correction r_n^2 is invisible on the larger Gaussian scale but is indispensable for the microscopic difference $R_n - J$. This explains why the earlier statement $J/R_n \rightarrow 1$ contains much less information.

8 Universality, perturbations, and what the constants mean

Dividing the equivalents in theorem 2.1 proves corollary 2.4. It is useful to separate the three mechanisms.

8.1 Finite-prefix universality and its exception

Suppose the slopes are eventually exactly quadratic. Every fixed λ_h , $h \geq 2$, influences lower-order terms in the microscopic row expansion, but none influences its leading equivalent. This is not because small parts are absent: the number of twos tends to infinity in probability. Rather, the large slope $\lambda_J \asymp J^2$ dominates the feedback sum so strongly that the numerical contribution of each fixed nonunit slope is negligible in its exponent.

The first slope is different. There are about n action-one factors, so their cumulative feedback contributes $\lambda_1 n^2/J^2 \sim \lambda_1 r_n^2$. For the baseline $\lambda_j = j^2$, one has $b = 2$. Keeping that tail but setting $\lambda_1 = c \geq 0$ gives

$$\frac{u_n^{(c)}}{u_n^{(1)}} \sim e^{(c-1)r_n^2}. \quad (8.1)$$

Thus equality of the tail of the defining slope sequence does not imply equality of coefficient equivalents unless the first slope is also fixed.

8.2 The linear tail is visible at a smaller exponential scale

For $\lambda_j = j^2 + \alpha j + o(j/\log j)$, the extra factor is $e^{\alpha r_n}$. The exact relation defining r_n rewrites it as

$$e^{\alpha r_n} = \left(\frac{n}{r_n(r_n + 1)} \right)^{\alpha/2}. \quad (8.2)$$

It is therefore polynomial in n , with a logarithmic correction. There is no need to re-center the leading Gaussian by an amount comparable to its standard deviation: the induced saddle shift is negligible on that scale, although the amplitude itself is not negligible.

8.3 A sharp constant-scale perturbation

A perturbation $\eta j / \log j$ produces the finite factor $e^{\eta/2}$, with the first slope and the linear coefficient held fixed. The mechanism is the exact evaluation

$$\frac{n - J}{J \log J} \sim \frac{r_n}{\log x_n} \longrightarrow \frac{1}{2}.$$

Any tail $o(j / \log j)$ disappears from the multiplicative equivalent. This gives an actual stability theorem, not a formal Taylor heuristic. For instance, changing the tail from j^2 to $j^2 + 2j / \log j$, while preserving λ_1 , multiplies the large-order coefficients asymptotically by e .

Remark 8.1 (Limits of this universality statement). The hypotheses deliberately specify the tail at the scale $j / \log j$. The weaker assumption $\lambda_j \sim j^2$ is not sufficient for the same equivalent: a perturbation such as $j^2 / \sqrt{\log j}$ is relatively small but is multiplied by a large exponent in the coefficient formula. Nor does this theorem cover sign cancellations or negative slopes.

9 A second saddle: the entire Borel transform

9.1 Entirety and the continuous coefficient envelope

The coefficient equivalent gives

$$\left(\frac{u_n}{n!} \right)^{1/n} \sim \frac{4}{(\log n)^2} \longrightarrow 0.$$

The root test proves that B in (2.15) is entire. For real $v > 0$, define $r = r(v)$ by $v = r(r+1)e^{2r}$, and let

$$F_0(v) = (2r+1) \frac{vr}{r+1}. \quad (9.1)$$

Implicit differentiation gives the exact identities

$$r'(v) = \frac{r(r+1)}{vD(r)}, \quad F'_0(v) = 2r. \quad (9.2)$$

Applying Stirling's formula to $n!$, the positive Borel summand is asymptotic, uniformly as the integer $n \rightarrow \infty$, to

$$e^{K(n;t)} c(n), \quad c(v) = \frac{\exp(br(v)^2 + \alpha r(v) + \eta/2)}{\sqrt{2\pi v D(r(v))}}, \quad (9.3)$$

where

$$\begin{aligned} K(v; t) &= F_0(v) - v \log v + v + v \log t \\ &= v \left[\frac{1}{r(v)+1} + \log \frac{t}{r(v)(r(v)+1)} \right]. \end{aligned} \quad (9.4)$$

The meaning of uniformity here is simple: the error in the coefficient equivalent is a sequence tending to zero, so its supremum over all $n \geq N$ tends to zero. Positivity permits this error to be passed through any tail sum.

9.2 Exact location and curvature of the Borel saddle

Equations (9.2) and (9.4) imply

$$K'(v; t) = \log \frac{t}{r(v)(r(v) + 1)}, \quad K''(v; t) = -\frac{2r(v) + 1}{vD(r(v))} < 0. \quad (9.5)$$

Let $s(s + 1) = t$. The unique critical point, its value, and its variance parameter are therefore

$$v_* = te^{2s}, \quad K(v_*; t) = se^{2s}, \quad \nu^2 = \frac{v_* D(s)}{2s + 1}. \quad (9.6)$$

In particular $\nu/v_* \rightarrow 0$. Uniformly for $v \in [v_*/2, 2v_*]$, one has $r(v) = s + O(1)$, and direct differentiation gives

$$|K'''(v; t)| = O(1/(v_*^2 s)), \quad |(\log c)'(v)| = O_\lambda(s/v_*). \quad (9.7)$$

After multiplication by ν^3 and ν , respectively, the right sides become $O(\sqrt{s/v_*})$ and $O_\lambda(s^{3/2}/\sqrt{v_*})$, both tending to zero. Consequently Taylor's formula and a discrete Gaussian comparison on this interval yield

$$\sum_{v_*/2 \leq n \leq 2v_*} e^{K(n; t)} c(n) \sim e^{K(v_*; t)} c(v_*) \sqrt{2\pi} \nu. \quad (9.8)$$

This is the same estimate as in [section 6](#), now with $\nu \rightarrow \infty$ and an even more slowly varying relative amplitude.

9.3 Tail control for the infinite positive sum

We record the tail argument because a saddle estimate on a finite window alone does not establish an entire-function growth theorem. On $[v_*/2, 2v_*]$, strict concavity loses at least cv_*/s at either endpoint. Also $K'(v_*/2; t) \geq c/s$ and $K'(2v_*; t) \leq -c/s$, for large s , by (9.5) or by integrating its second derivative. For $n \leq v_*/2$, the core exponent is bounded by its endpoint value, since it increases up to v_* . The amplitude has logarithm bounded above by $O_\lambda(1 + s^2)$ on $1 \leq n \leq v_*/2$. Even after multiplication by the number of terms, this tail is negligible compared with the central sum.

For $v \geq 2v_*$, concavity gives

$$K(v; t) \leq K(2v_*; t) - \frac{c}{s}(v - 2v_*).$$

From (9.2), $|(\log c)'(v)| \leq C_\lambda(1 + \log v)/v$ for large v . The supremum of this bound over $v \geq 2v_*$ is $O_\lambda(s/v_*) = o(1/s)$. Thus half of the displayed linear decay absorbs all possible amplitude growth. The remaining tail is a geometric sum with factor of order s , multiplied by an endpoint already smaller by $e^{-cv_*/s}$. Finitely many initial Borel terms grow only polynomially in t , and are negligible. These bounds also justify insertion of the uniform coefficient equivalent into the infinite tail sum.

Substitution of (9.6) into (9.8) now gives

$$e^{se^{2s}} \frac{e^{bs^2 + \alpha s + \eta/2}}{\sqrt{2\pi v_* D(s)}} \sqrt{\frac{2\pi v_* D(s)}{2s + 1}},$$

which is exactly (2.16). This proves [theorem 2.5](#).

9.4 Consequences and directional limitations

Since $s = \sqrt{t} - \frac{1}{2} + O(t^{-1/2})$,

$$\log B(t) \sim se^{2s} \sim \frac{\sqrt{t}}{e} e^{2\sqrt{t}}, \quad \log \log B(t) \sim 2\sqrt{t}. \quad (9.9)$$

The first equivalent and the full formula retain information lost by the second. With fixed tail parameters, changing λ_1 by Δ changes $B(t)$ by a multiplicative factor asymptotic to $e^{\Delta s^2}$, although it leaves the leading equivalent for $\log B$ unchanged.

For every $q > 0$,

$$\frac{1}{q} \int_0^\infty e^{-t/q} B(t) dt = +\infty. \quad (9.10)$$

Indeed, (9.9) gives $\log B(t) - t/q \rightarrow +\infty$; the positive integrand does not even tend to zero. An entire Borel transform does not automatically have the growth bounds needed for directional Borel–Laplace summation; this distinction is part of the standard summability framework [5]. Nothing in this positive-axis proof determines growth on the negative axis, angular cancellation, or a Stokes constant.

There is also a direct obstruction to a positive-real literal solution of (2.1). If $q \in (0, 1)$ and $u > 0$, then $j \log q + \lambda_j u \rightarrow +\infty$, so $q^j e^{\lambda_j u}$ fails to tend to zero. Any proposed accelerated positive-direction value must therefore specify a continuation or regularization of the kernel; it cannot silently be called a solution of that divergent literal sum.

10 Least-term asymptotics without an unwarranted remainder claim

For each fixed $t > 0$, theorem 2.1 implies $u_n^{1/n} \rightarrow \infty$. Thus $u_n t^n \rightarrow \infty$, and its minimum is attained at a finite index. Put $R = \frac{1}{2} \log(1/t)$ and use the continuous envelope

$$Q(v; t) = F_0(v) + v \log t + g(r(v)), \quad g(r) = br^2 + \alpha r + \eta/2 - \frac{1}{2} \log D(r). \quad (10.1)$$

By (9.2), the derivative of the core $F_0(v) + v \log t$ is $2r(v) - 2R$. Its unique minimum is at $N_0 = R(R+1)e^{2R} = R(R+1)/t$, where the core value is exactly $-R^2/t$.

On a fixed relative neighborhood of N_0 , the core second derivative is $2r'(v) \asymp 1/N_0$, while

$$\frac{\partial}{\partial v} g(r(v)) = O_\lambda(R/N_0), \quad \frac{\partial^2}{\partial v^2} g(r(v)) = O_\lambda(R/N_0^2).$$

It follows by the mean value theorem that the minimum of the smooth full envelope is displaced from N_0 by $O_\lambda(R)$. The resulting change in its minimum value is $O_\lambda(R^2/N_0) = O_\lambda(t) = o(1)$. Rounding either this minimum or N_0 to an integer changes the value by $o(1)$.

For clarity, the local minimum cannot be bypassed at a far-away index. The full envelope is strictly convex for all sufficiently large v : its positive core curvature is asymptotic to $1/v$, and the amplitude curvature is $O_\lambda((\log v)/v^2)$. Its derivative has opposite signs at $N_0/2$ and $2N_0$ when t is small. Therefore it has one eventual minimum. On the complement of any fixed relative neighborhood of N_0 , integration of the core curvature gives a loss of order N_0 , whereas the amplitude variation is only $O_\lambda(R^2)$ on fixed relative windows; convexity continues that loss beyond them. The finitely many smaller indices give terms of polynomial size in t , much larger than $e^{-R^2/t + O(R^2)}$.

The coefficient equivalent means that $\log(u_n t^n) = Q(n; t) + o(1)$, uniformly over all sufficiently large integers. Thus it may be minimized without losing more than $o(1)$ in the minimum

value. Every minimizing index tends to infinity and must satisfy $n/N_0 \rightarrow 1$, by the preceding convexity loss. Evaluating g at $r = R$ and exponentiating proves [theorem 2.6](#), including its assertion about a nearest integer to N_0 .

Remark 10.1 (A scale, not yet an analytic error theorem). The dominant least-term scale is

$$\exp\left(-\frac{\log^2(1/t)}{4t}\right). \quad (10.2)$$

It is smaller than $e^{-A/t}$ for every fixed $A > 0$, as $t \downarrow 0$. This does not imply that a sectorial solution has a remainder of this size. A remainder theorem must relate the actual function to its truncated Taylor series and control cancellations and analytic continuation. The source article provides a left-sector fixed point, but its Poincaré statement alone supplies no such uniform optimal-order estimate [\[1\]](#).

11 Exact computation and reproducibility

The accompanying program is designed to audit finite algebra and to show how slowly the asymptotic regime can be approached. Its exact calculations are logically separate from the asymptotic proofs.

11.1 An integral coefficient recurrence

For integer slopes, let

$$b_n = n!u_n, \quad E_{j,m} = m![q^m]e^{\lambda_j \widehat{U}(q)}. \quad (11.1)$$

Differentiation of the exponential identity gives

$$\begin{aligned} E_{j,0} &= 1, \\ E_{j,m} &= \lambda_j \sum_{k=1}^m \binom{m-1}{k-1} b_k E_{j,m-k}, \quad m \geq 1, \end{aligned} \quad (11.2)$$

$$b_n = \sum_{j=1}^n \frac{n!}{(n-j)!} E_{j,n-j}. \quad (11.3)$$

When computing b_n , the required exponential coefficients have $m \leq n-1$, so only previously computed b_k are used. Starting from $E_{j,0} = 1$, every operation in these formulas is an integer operation. This gives an independent implementation of the formal fixed point without enumerating partitions.

A second implementation enumerates integer partitions of n and sums [\(3.2\)](#) using exact rational arithmetic. Agreement of these two implementations is a useful test because their combinatorial organization and normalization differ substantially.

11.2 Executed checks

The program computed exact coefficients through degree 256 for four models: the unmodified quadratic slopes, and the three modifications $\lambda_1 = 0$, $\lambda_1 = 3$, and $\lambda_2 = 7$, with all other slopes unchanged. For each model, all degrees 1 through 18 were also computed by independent partition enumeration. All 72 exact comparisons passed.

For the quadratic model it additionally summed the exact weights [\(5.8\)](#) over all possible $J \geq 3$ and d . The resulting mass was checked to lie between zero and the full coefficient. The

reported probabilities are evaluations of ratios of these exact integers. The asymptotic formula, logarithms, and displayed ratios were evaluated at 80-decimal working precision using `mpmath`; they are not outward-rounded interval bounds.

Table 1: Quadratic slopes: the exact coefficient divided by the leading equivalent, and the probability of one large action with only ones and twos.

n	r_n	u_n/F_n	Good-configuration probability
20	1.087760	3.531186	0.561985
40	1.298005	2.090003	0.767695
80	1.519691	1.689076	0.849825
120	1.654076	1.557336	0.876061
180	1.791611	1.455860	0.896387
256	1.913461	1.384043	0.910996

At degree 256, the exact coefficient is still about 38% larger than the leading equivalent F_n . The theorem is an asymptotic statement, and its conservative $O((\log n)^2/\sqrt{n})$ rate is not a small numerical bound at those orders. The observed values support neither a monotonic-convergence claim nor an explicit finite-order error constant. Those would require separate arguments.

Table 2: Finite-perturbation ratios. Each displayed column tends to one by [corollary 2.4](#); the finite values are diagnostics, not proofs.

n	$u_n^{[\lambda_2=7]}/u_n$	$e^{-2r_n^2}u_n^{[\lambda_1=3]}/u_n$	$e^{r_n^2}u_n^{[\lambda_1=0]}/u_n$
20	3.642423	6.041575	0.888086
40	1.897724	2.674726	0.960560
80	1.538280	1.699454	0.996221
120	1.436076	1.516262	1.005374
180	1.358496	1.394877	1.010625
256	1.303662	1.317723	1.013079

11.3 Files and rebuild procedure

The package contains the article source and PDF, the verification program, exact coefficient tables, numerical diagnostics, a machine-readable check report, source-provenance notes, and a build script. From the package root, run

```
python -m pip install -r requirements.txt
python code/verify.py -order 256 -check-order 18
sh build.sh
```

The exact recurrence and partition calculations use Python’s standard library. The listed `mpmath` dependency is used for diagnostics. The PDF needs a standard LaTeX installation with the packages named in the source. No internet retrieval, external figure, or external bibliography file is required to rebuild it.

12 Proof status and a feasible formalization route

Every main assertion above has a conventional mathematical proof in this article. None has been machine-checked in Lean in this work. The finite program checks do not certify infinite tails, limiting distributions, or Borel growth.

A useful formalization can be staged without implementing a complete transseries field. First formalize the coefficientwise contraction and the finite coefficient identity in a formal-power-series ring. The next stage is finite: multiplicity vectors, the marked-largest-part inequality, the exact ones/twos weights, and the positive exponential majorant. These are natural targets for algebraic and combinatorial proof automation.

The central analytic interface should be the uniform row lemma, with its parameters and macroscopic window made explicit. A reusable discrete Laplace theorem should accept bounds on the second and third derivatives, a bound on the logarithmic derivative of the amplitude, and a certified outside-window estimate. The proof in [section 6](#) identifies exactly those hypotheses. Such an interface would also support the second Borel saddle, rather than requiring a special-purpose proof for each sum.

Finally, probability statements should be formulated in total variation before passing to weak limits. The Poisson coupling and the uniform pointwise bounds are simpler, more informative intermediate goals than an isolated central limit theorem. The distinction between coefficient asymptotics and analytic remainders should remain explicit in the API; there is no valid implication from [theorem 2.6](#) alone to a remainder certificate.

Result	Dependencies and status
Positive coefficient identity	Classical Lagrange inversion, with a finite coefficientwise specialization; rederived here and checked at finite orders.
Coarse localization	Positive envelope, Stirling estimates, one-dimensional strict concavity; rederived from the source article’s mechanism.
Microscopic row lemma	Marking, exponential majorization, a small-part tail bound, and exact ones/twos weights; proved here.
Full coefficient equivalent	Microscopic row lemma, explicit saddle, Gaussian sum, and perturbation expansion; proved here.
Joint and local limits	Pointwise row weights, summed absolute saddle error, Poisson coupling, and Stirling; proved here.
Borel equivalent	Coefficient equivalent, a second saddle, and control of both infinite tails; proved here.
Least term	Coefficient equivalent, convex minimization, integer rounding; proved here, but not an analytic remainder theorem.

13 Further research questions

The following questions continue the proved results rather than restating questions already settled in this article. Assertions labeled as heuristics are not theorems of this paper.

13.1 All-order corrections and the first appearance of each small slope

For eventually quadratic slopes, determine a systematic expansion of u_n/F_n in inverse powers of n , with coefficients expressed in r_n and the fixed initial slopes. The exact microscopic formula provides a route: expand the falling factorial, the feedback logarithm, and the discrete saddle to matching orders. The first omitted primitive cloud has scale $m^3/\Lambda^2 \asymp (\log n)^4/n$. At what precise order does each λ_h , $h \geq 2$, first become visible, and which coefficients admit

sign or monotonicity statements? A useful answer should include a uniform truncation bound, not only a list of formal correction polynomials.

13.2 The cloud hierarchy for nonquadratic slopes

For $\lambda_j \sim aj^p$, $p > 1$, the source article proves only macroscopic concentration and logarithmic coefficient growth. At a one-large-action configuration, the natural formal small-part activity is approximately

$$\mu_h \approx \frac{n^h}{(aJ^p)^{h-1}}, \quad J \asymp n/\log n.$$

Thus its power of n is $p - (p - 1)h$. This is a heuristic predicting changes at $p = h/(h - 1)$: for $p > 2$, even the twos should disappear; for $1 < p < 2$, several small sizes may occur in large numbers. At a critical exponent the logarithmic factors still matter. Prove the appropriate multi-cloud limit theorem, including interactions and the correct shifted saddle. The present quadratic theorem proves one member of that hierarchy, not the hierarchy itself.

Editorial note (ProveIt, 2026-09-29). Partly answered for $\lambda_j = aj^p$ (after finitely many nonnegative changes) by two other packages of this series. `Finite_Core_Universality_Exponential_Feedback/` proves that an analytic core of M actions gives coefficients asymptotic to u_n exactly when $M(p - 1) > 1$ (its Theorem `thm:cutoff`), so for $p > 2$ a one-action core suffices, as the heuristic above predicts, with a multiplicative saddle equivalent (its Theorem `thm:saddle`). `Poisson_Layers_Finite_Core_Boundary/` proves Poisson and Gaussian laws for the number of occurrences of the first omitted action, jointly with the largest action, throughout $(M + 1)(p - 1) > 1$ (its Theorems `thm:probability` and `thm:joint`). The joint law of the whole cloud vector is not treated.

13.3 Boundary perturbations without a smooth asymptotic tail

The scale $j/\log j$ is now known to affect the constant in the coefficient equivalent. Classify tails of the form $\lambda_j = j^2 + \alpha j + (j/\log j)\eta_j$, where η_j oscillates instead of converging. Slow oscillations suggest an amplitude sampled near $j = x_n$; oscillations across a window of width σ_n may instead be averaged by a Gaussian profile. Determine necessary and sufficient local regularity hypotheses, and construct examples with genuinely nonconvergent normalized coefficients.

13.4 Compositional inverses and cancellation-sensitive large order

Let $\widehat{Q} = \widehat{U}^{(-1)}$. The source article transfers Gevrey-class membership but does not determine exact type or a coefficient equivalent. Does the one-large-action mechanism survive the signed reversion weights after cancellation? Determine the first exponential and subexponential scales of the inverse coefficients, and identify when the first-slope factor remains visible. Positivity of (3.2) cannot simply be reused for a signed inversion formula.

Editorial note (ProveIt, 2026-09-29). The exact type is answered by `Sharp_Weighted_Type_Formal_Reversion/`: reversion preserves the Gevrey type (its Theorem `thm:inversefeedback`) and the refined inverse type is 4 for $\lambda_j = j^2$ (its Corollary `cor:quadratic`). A coefficient equivalent for $\lambda_j = aj^2$ is claimed, independently and not yet reviewed, by `Quadratic_Exponential_Feedback_After_Reversion/` and `Signed_Quadratic_Feedback_Inversion/` (Theorem `thm:main` in each):

$v_n \sim -S_n \exp(-(1 + 1/a)r_n)$ in the notation of the finite-core package; the first of them also treats slope tails $aj^2 + dj + e$ and $\kappa j / \log j$ perturbations, and in both the first-slope factor enters through $\lambda_1 + w_2$.

13.5 Angular growth and negative-direction Borel summation

The positive-axis Borel equivalent is now explicit, but it gives no cancellation estimate for $B(-t)$ or for nearby complex rays. Determine sectorial growth, possible indicator transitions, and whether a suitable negative-direction Borel–Laplace sum agrees with the literal left-sector fixed point. The proof must establish the analytic identification as well as growth; an entire Borel transform alone settles neither.

Editorial note (ProveIt, 2026-09-29). Answered in two senses by later packages. `Negative_Ray_Summability_Exponential_Feedback/` (its Theorem `thm:threshold`) proves, for unit amplitudes and $\lambda_j = O((j \log j)^2)$, which contains (Q), fine (fixed-width half-strip) Borel–Laplace summability of $\widehat{U}$ and of its inverse in direction π , with sums that satisfy the literal kernel near the negative axis. `Natural_Boundaries_Quadratic_Exponential_Feedback/` proves that for $\lambda_j = j^2$ (its Theorem `thm:main`), and for slopes that are eventually $j^2 + \beta j + \gamma$ (its Theorem `thm:rational-main`), no open arc of directions around π gives angular 1-summability, because the literal inverse has a natural boundary on the imaginary diameter.

13.6 Acceleration adapted to the double-exponential growth

Construct and analyze an acceleration procedure adapted to $\log B(t) \sim (\sqrt{t}/e)e^{2\sqrt{t}}$. Specify the analytic continuation or regularization of the countable kernel that the proposed sum solves. Different choices may be inequivalent, since the literal positive-real kernel diverges for every positive candidate value. A complete result should identify uniqueness conditions and compatibility with composition or inversion.

13.7 From the least term to the actual sectorial remainder

The proved least-term scale (10.2) suggests a refined optimal-order problem for the left-sector analytic solution. Does its Taylor remainder at order near $R(R+1)/t$ have that scale, a different scale, or additional oscillatory cancellation? Establish a uniform remainder theorem on proper sectors, and only then compare it with the least term. This is a sharper question than coefficient Gevrey membership.

Editorial note (ProveIt, 2026-09-29). The uniform remainder theorem is supplied by `Sharp_Negative_Direction_Summability_Exponential_Feedback/`: strong Gevrey-1 remainders on every proper left sector (its Theorem `thm:headline`) and an actual truncation error at most $C \exp[-c \log^2(1/|q|)/|q|]$ at an index of order $\log^2(1/|q|)/|q|$ (its Corollary `cor:actualerror`), with an unspecified constant c ; the negative-ray package bounds the inverse on the same scale (its Theorem `thm:optimal`, an upper bound). Whether the actual remainder has the scale of the least term remains open.

13.8 Amplitudes, noninteger actions, and several feedback components

Replace the model by

$$U = \sum_j c_j q^{a_j} e^{\lambda_j U}, \quad c_j \geq 0,$$

first with integer actions and then with well-based real actions. Which amplitude and action assumptions preserve a one-large-action reduction? The unit-action reservoir is essential to the present balance. For vector feedback, determine whether an analogous reduction is governed by a nonnegative matrix or by several competing large components. Signed or complex amplitudes require a separate noncancellation theory.

13.9 Explicit finite certificates and scalable computation

The current relative error constants are asymptotic and depend on the fixed exceptional slopes. Extract computable thresholds and constants from the microscopic and saddle bounds, use directed interval arithmetic, and give finite enclosures for u_n at orders beyond exact recurrence computation. A particularly useful tool would certify both the coefficient and the probability of exceptional configurations, with its proof obligations structured for eventual Lean verification.

14 Conclusion

The unresolved finer-asymptotics question in the repository's exponential-feedback article has a concrete answer for the class (Q). The coefficient is governed by one explicit scalar saddle, a Gaussian large action, and a Poisson cloud of twos. The cloud supplies a factor $e^{r_n^2}$ that would be missed by retaining only a large action and unit actions. The unit-action feedback supplies the additional $e^{\lambda_1 r_n^2}$, explaining both finite-prefix universality and its sharp exception.

The resulting equivalent identifies a hierarchy of perturbation scales, strengthens Borel growth from double-logarithmic precision to a full multiplicative formula, and determines the least-term scale. These results are proved for a precisely specified model and tail class. They do not settle unrestricted transseries summability, signed reversion, or the optimal remainder of a sectorial realization. Those remaining questions now have more precise parameters and sharper candidate scales.

A Notation and proof-audit checkpoints

Symbol	Meaning
u_n	Coefficient of the formal fixed point, without factorial scaling.
b	The fixed number $1 + \lambda_1 = u_2$.
b_n	The separate computational quantity $n!u_n$.
K, J, D_2	Number of factors, largest action, number of twos.
R_n	The excess-action coordinate $n - K + 1$.
r_n	Positive solution of $r_n(r_n + 1)e^{2r_n} = n$.
x_n, m_n	Saddle large action and remaining unit mass, $n/(r_n + 1)$ and $nr_n/(r_n + 1)$.
$D(r), \sigma_n^2$	$2r^2 + 4r + 1$ and $m_n/D(r_n)$.
A_J	Row Poisson mean $(n - J)^2/\lambda_J$.
s	Borel saddle parameter, $s(s + 1) = t$.
R	Least-term parameter $\frac{1}{2} \log(1/t)$, not R_n .

An independent proof audit can focus on five points. First, verify the marking factor in (5.3): multiple largest parts must be *overcounted*, not treated as exact. Second, keep the factor e^{μ_2} ; discarding the entropy of the twos loses a diverging multiplicative factor. Third, check that the unspecified tail in (Q) yields a *uniform* $o(1)$ in (6.14), without assuming smoothness. Fourth, distinguish total variation from rescaled pointwise local limits; the extra bound in corollary 7.1 is necessary. Fifth, keep the positive-axis Borel estimate and the least-term minimization separate from an analytic remainder theorem.

References

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